

RELATORIO DE ANALISE - MODELO VARMAX

Hiperparametros do Modelo:

```
p      (endog_lag)           : 2
x_lag (exog_lag)           : 2
r      (exog_real_degree)   : 3
i      (exog_img_degree)    : 3
```

Equacao do Modelo:

$$Y(t) = AR(1..2) + X_{real}^{(1..3)} + X_{img}^{(1..3)} \text{ com lag}(1..2)$$

Base de Dados:

```
Tamanho           : 48383
N° Replicações    : 1
Randomizada       : False
Seed              : 42
```

Resumo Estatístico:

Statespace Model Results

Dep. Variable:	['Yreal', 'Yimg']	No. Observations:	290298
Model:	VARX(2)	Log Likelihood	-459933.399
	+ intercept	AIC	919964.798
Date:	Sat, 09 May 2026	BIC	920483.153
Time:	14:58:20	HQIC	920114.934
Sample:	0		
	- 290298		
Covariance Type:	opg		

METRICAS DO MODELO

Hiperparametros: $p=2$, $x_lag=2$, $r=3$, $i=3$

Metricas Gerais:

RMSE Medio	: 0.532619
AIC	: 919964.7982
BIC	: 920483.1527

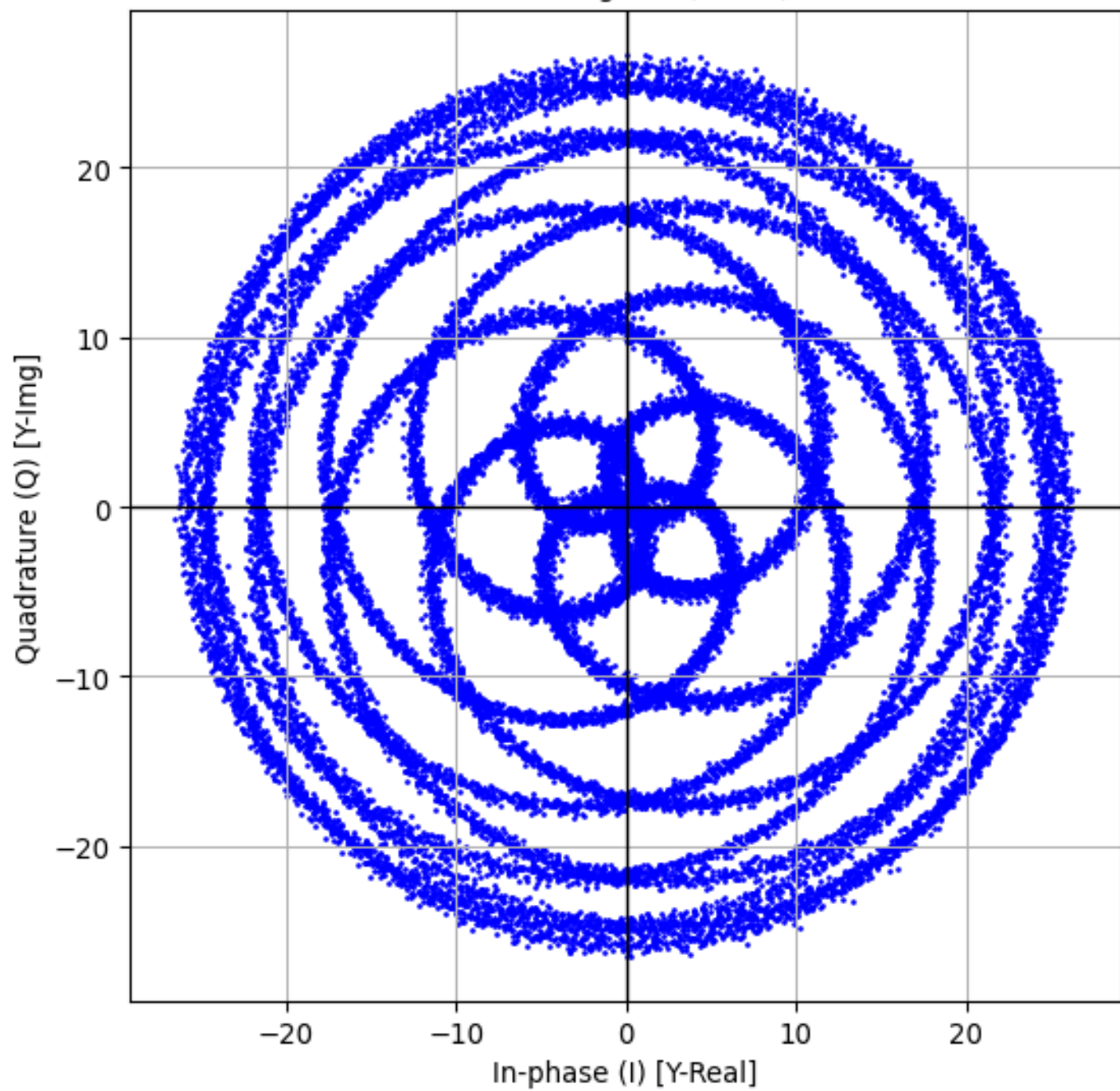
Componente Real (Yreal):

MSE	: 0.283116
RMSE	: 0.532087
MAE	: 0.439994
R2	: 0.998332
Adj R2	: 0.998332
STD	: 0.532087

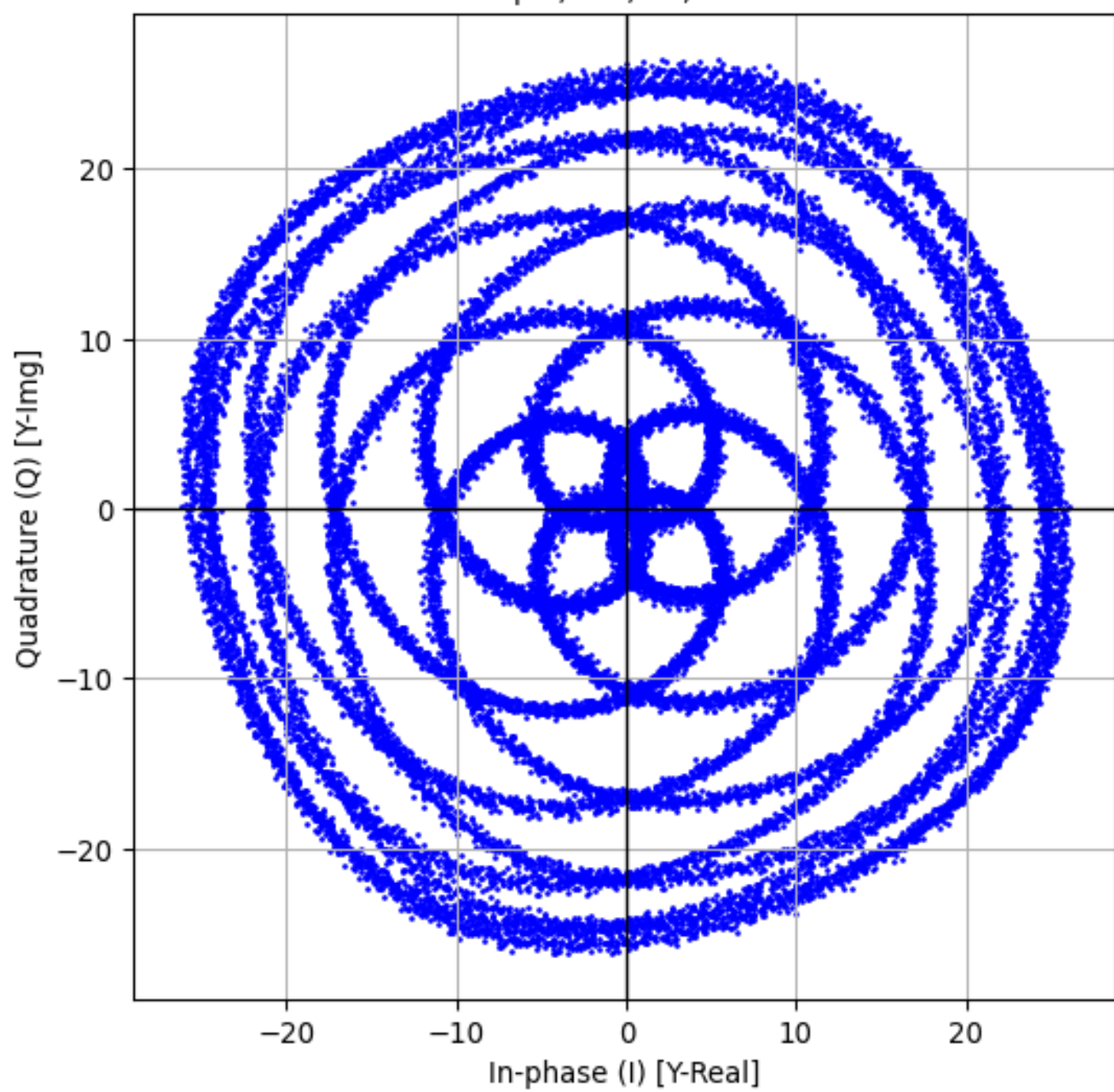
Componente Imaginaria (Yimg):

MSE	: 0.284251
RMSE	: 0.533152
MAE	: 0.441164
R2	: 0.998321
Adj R2	: 0.998321
STD	: 0.533149

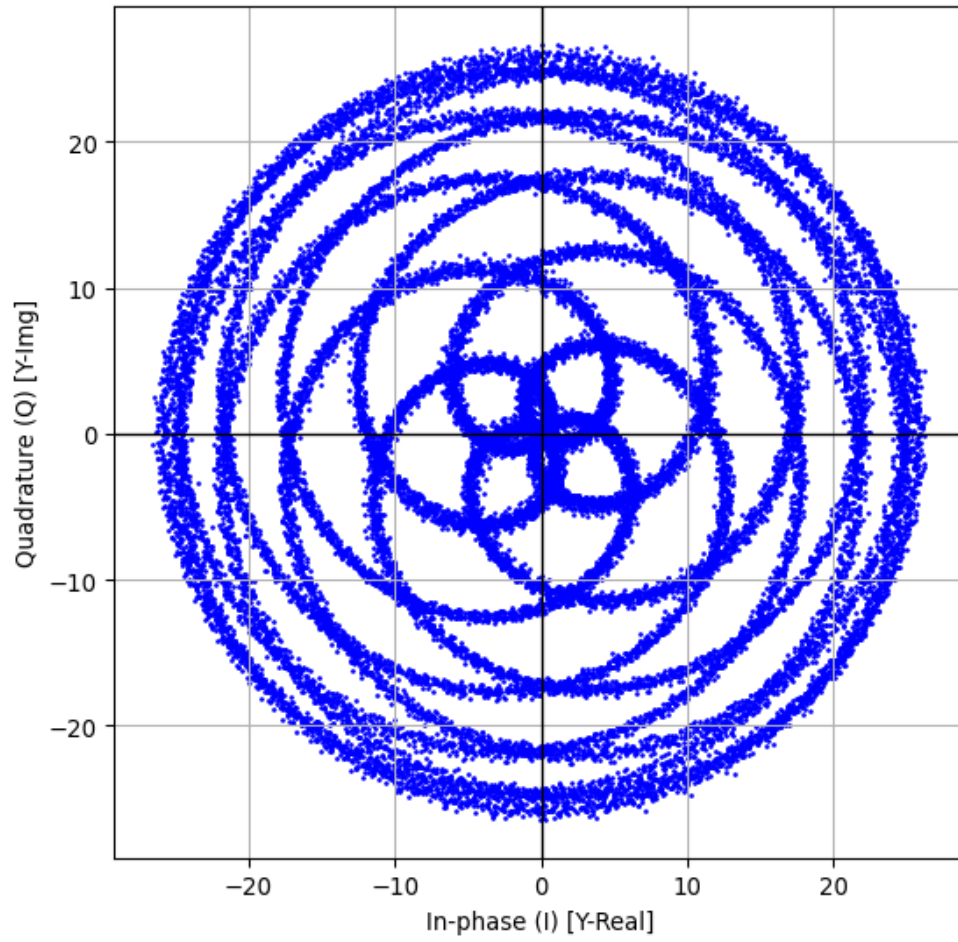
Dados Originais (Treino)



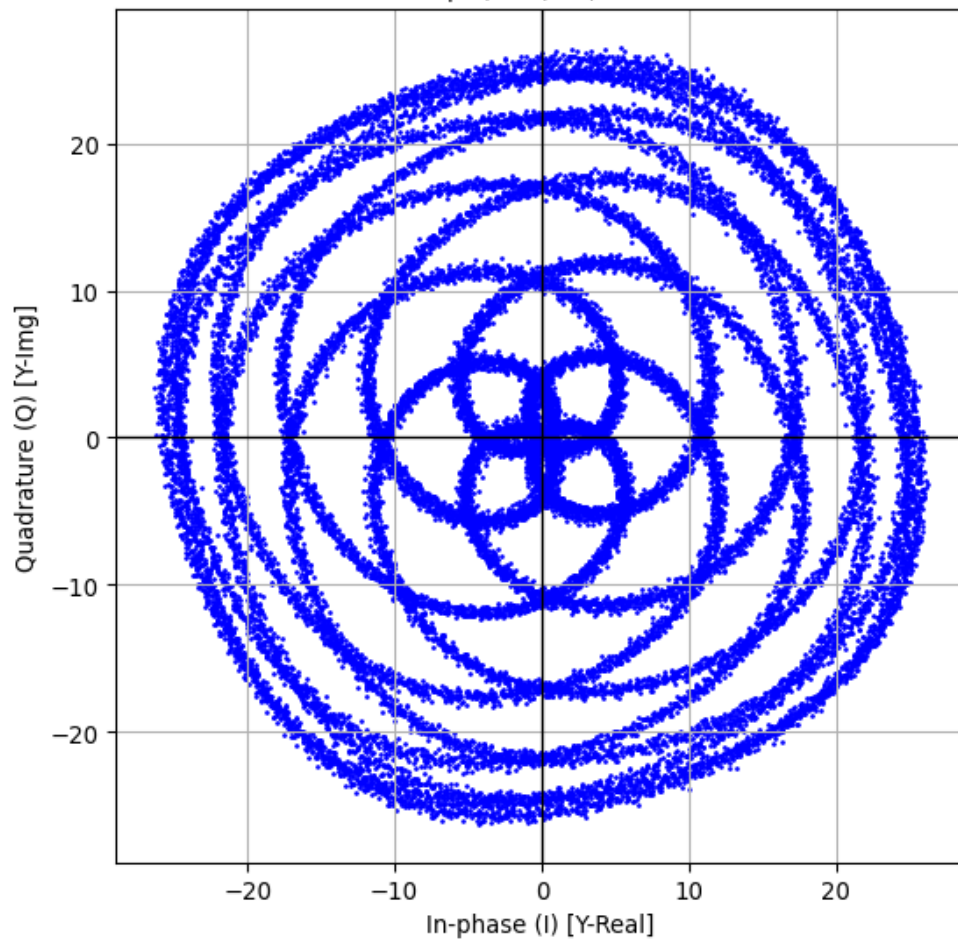
Forecast (Treino)
p:2, xl:2, r:3, i:3



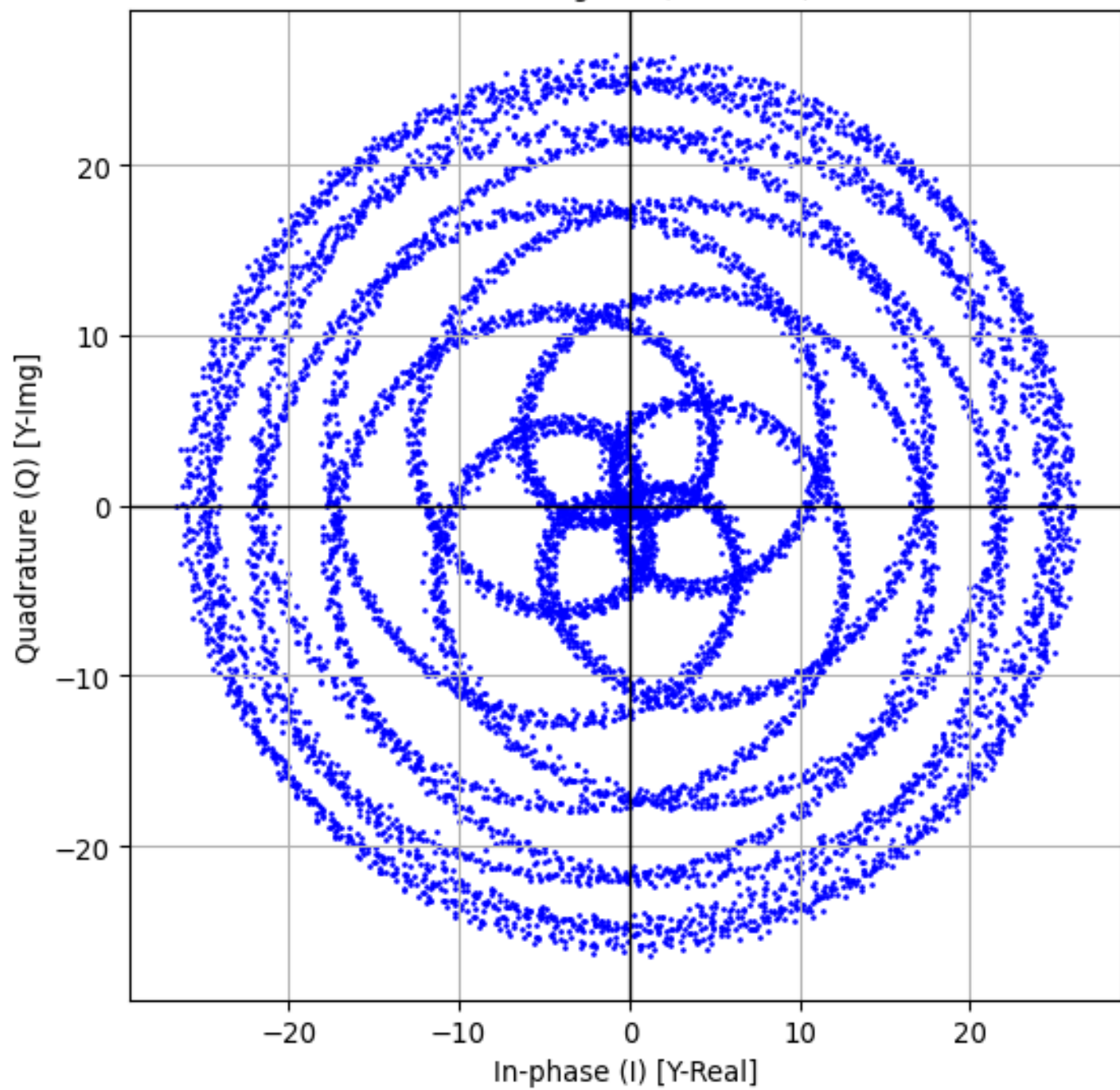
Dados Originais (Treino)



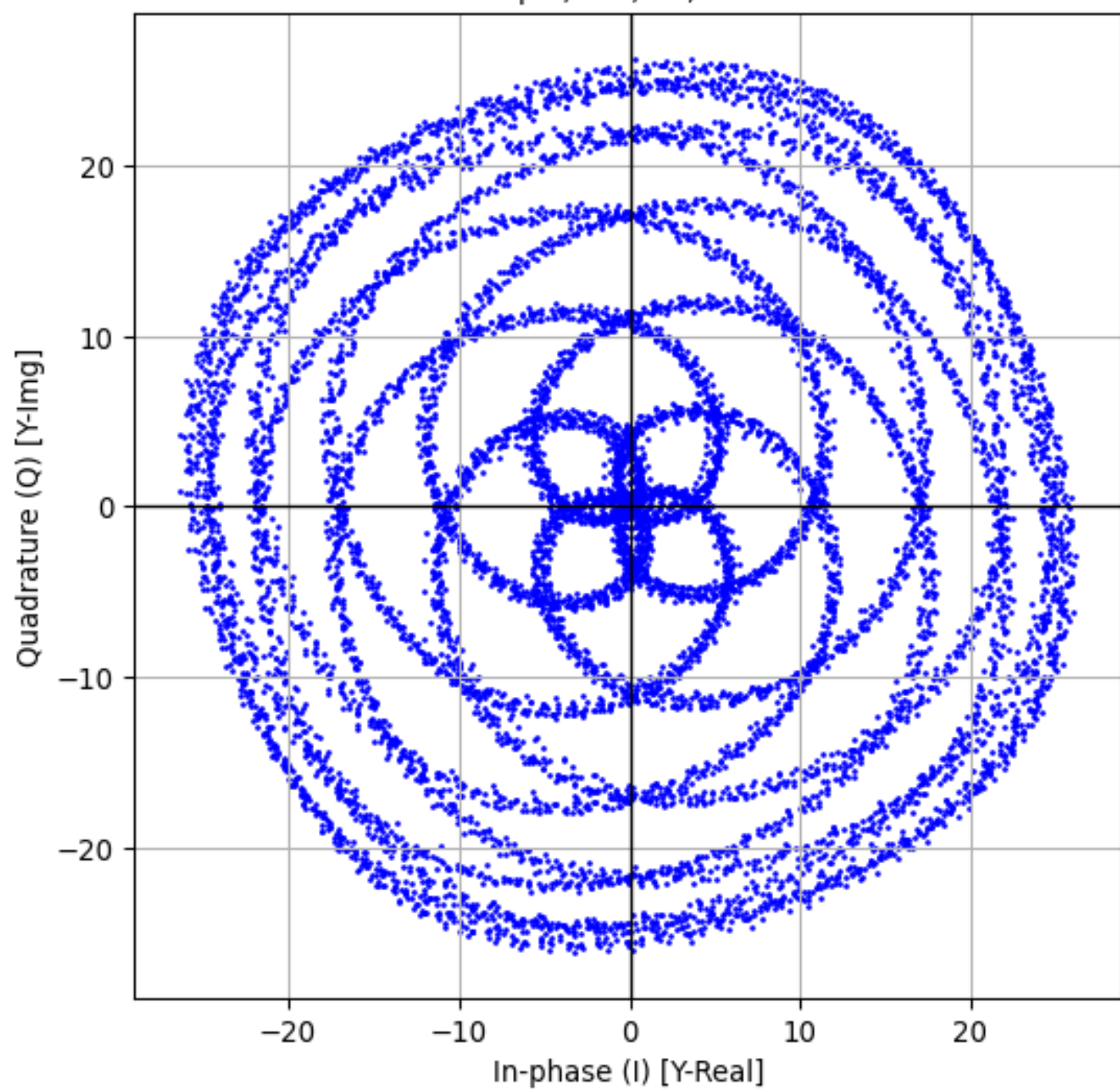
Forecast (Treino)
p:2, xl:2, r:3, i:3

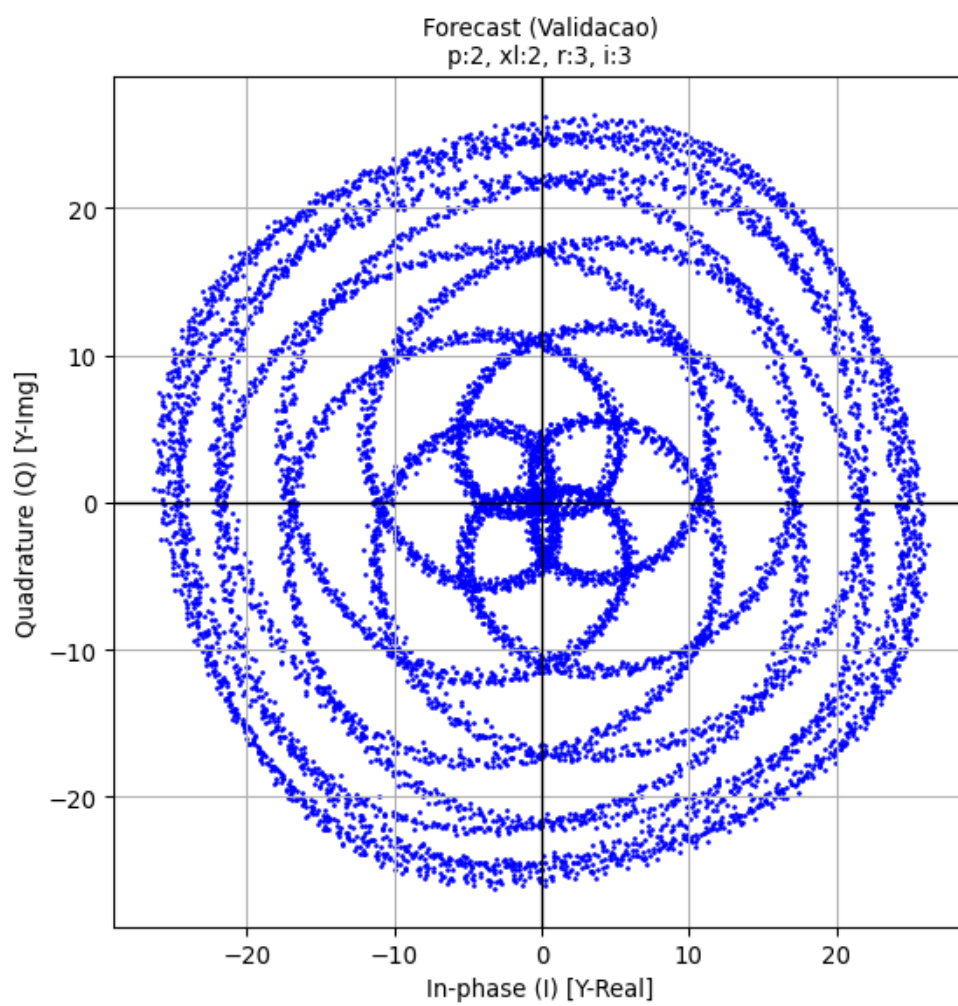
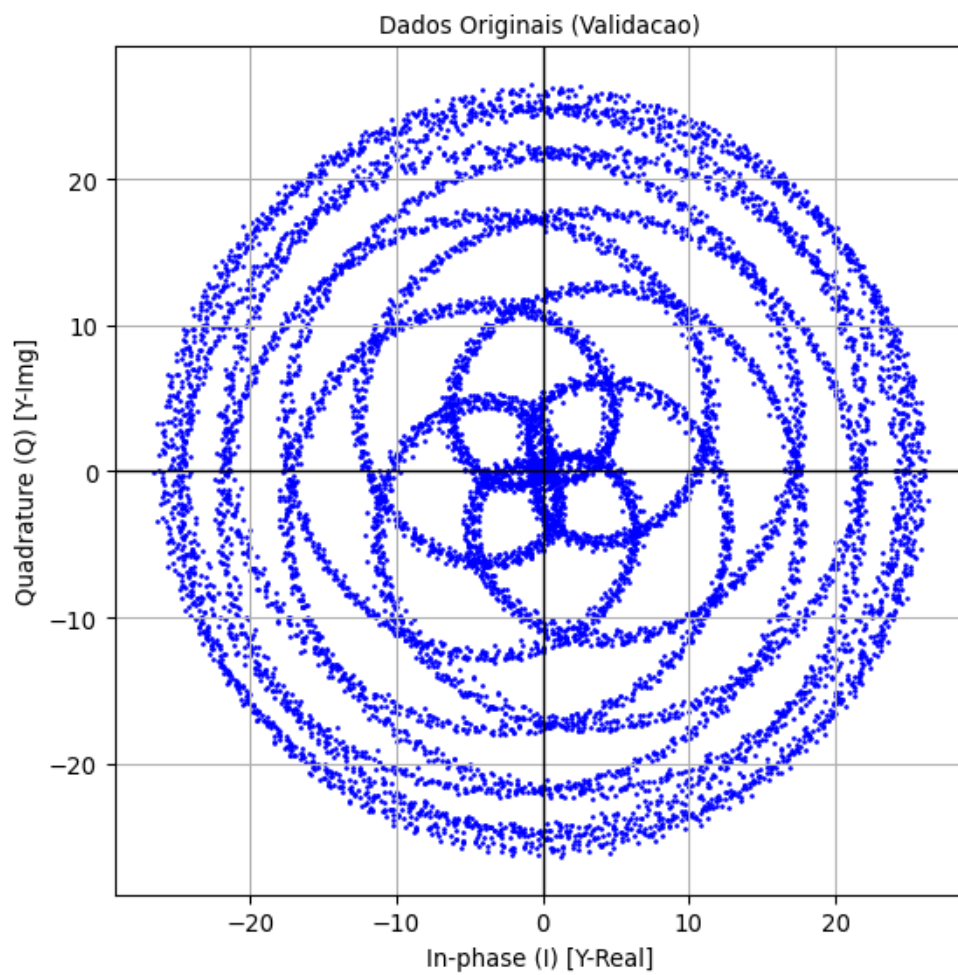


Dados Originais (Validacao)

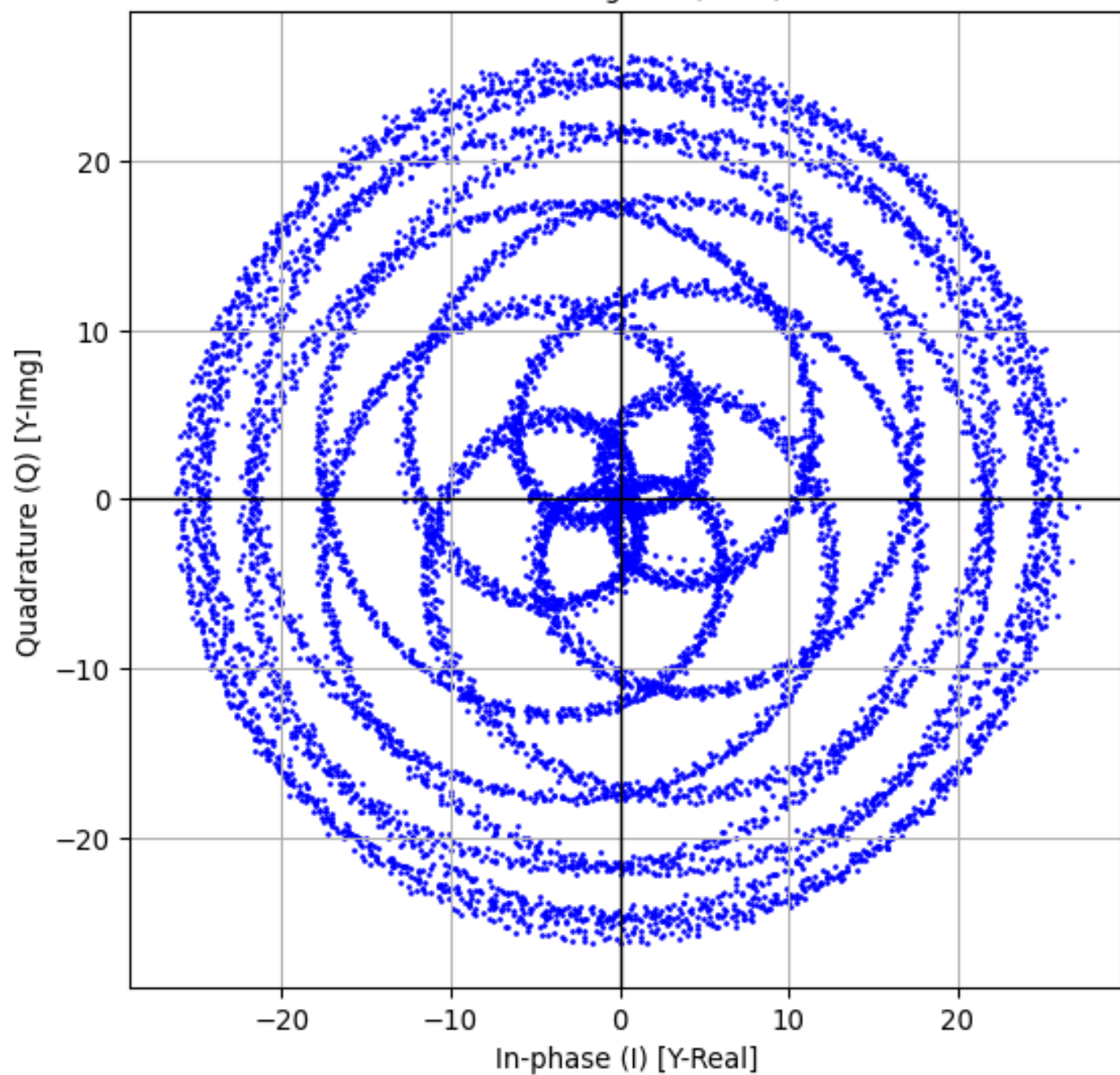


Forecast (Validacao)
p:2, xl:2, r:3, i:3

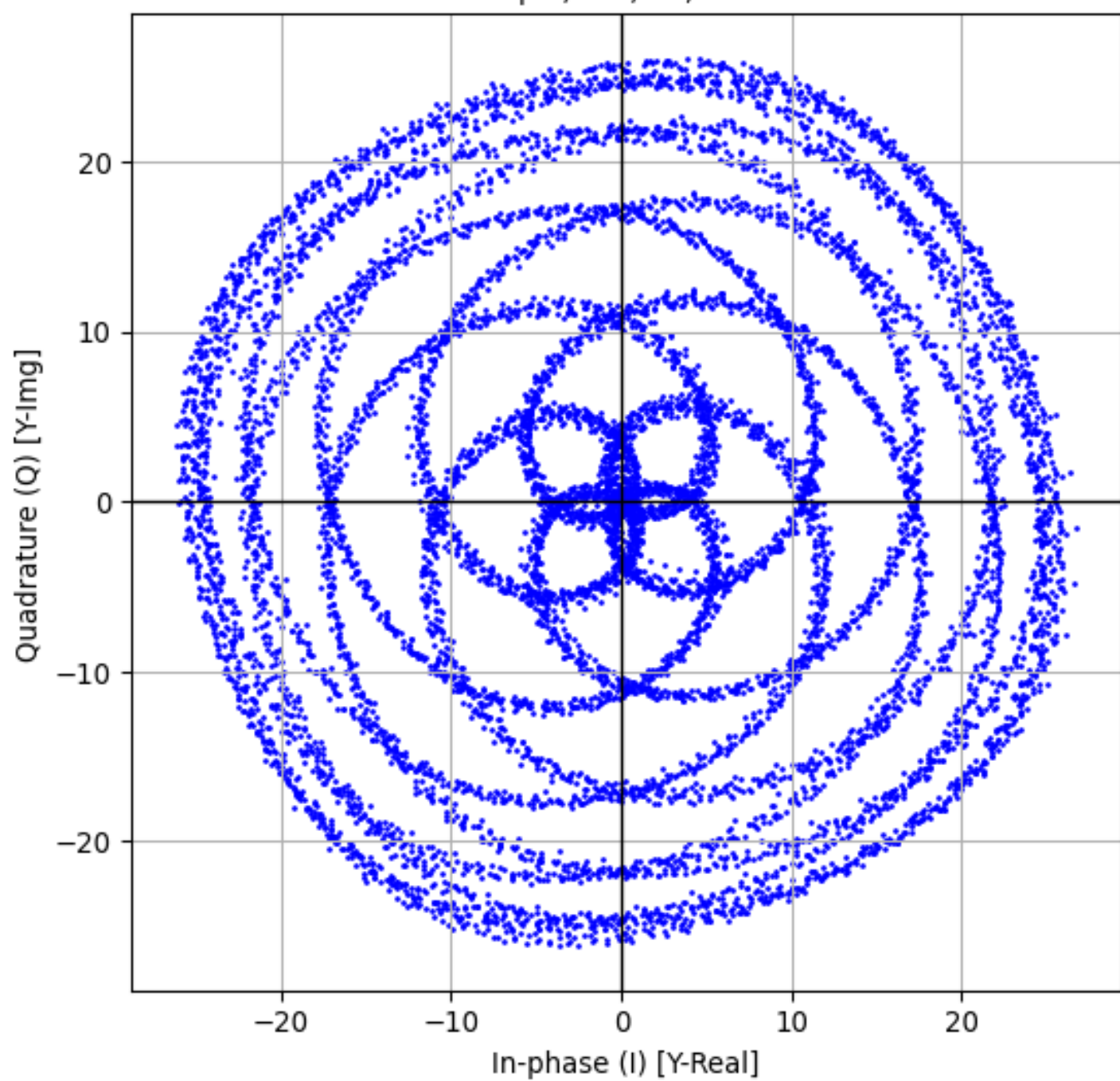




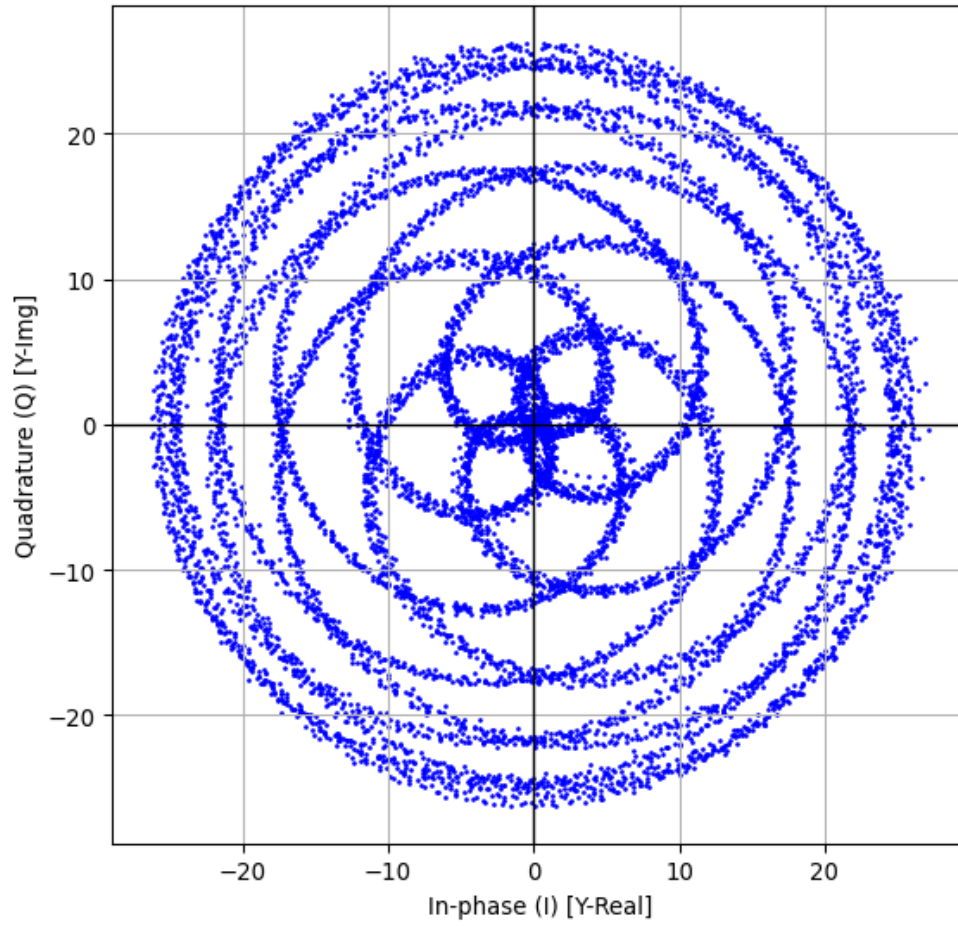
Dados Originais (Teste)



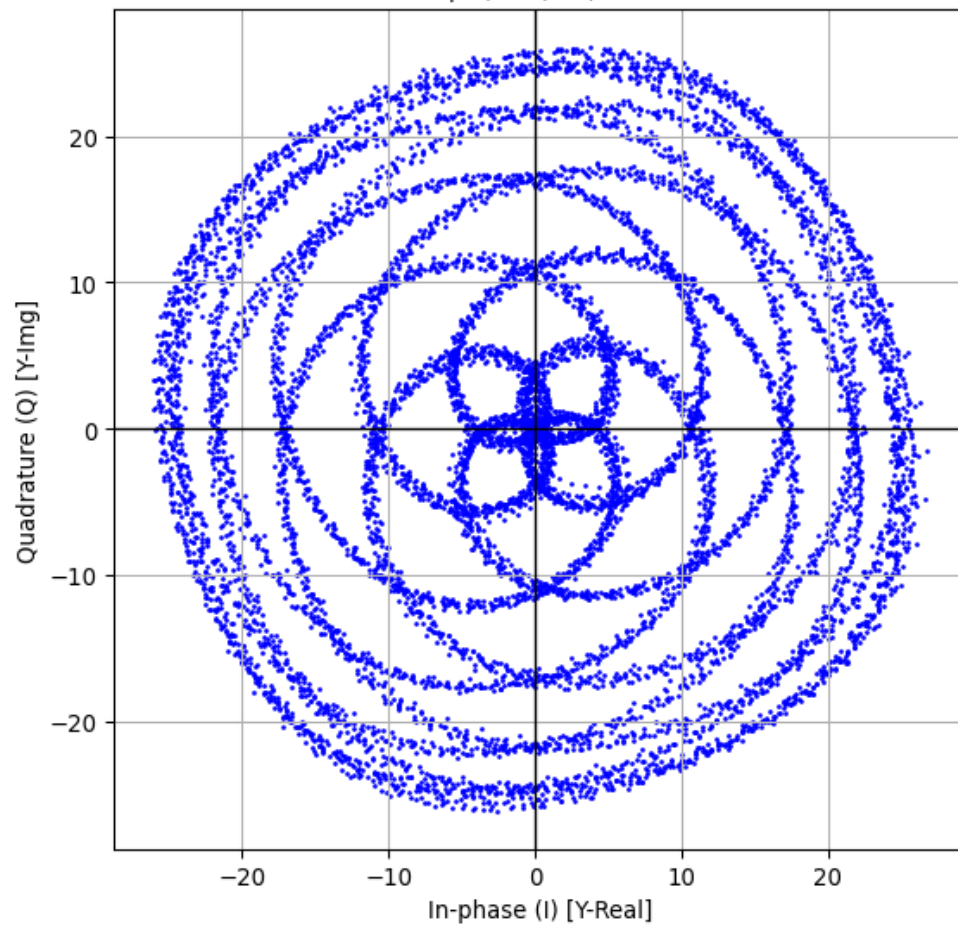
Forecast (Teste)
p:2, xl:2, r:3, i:3



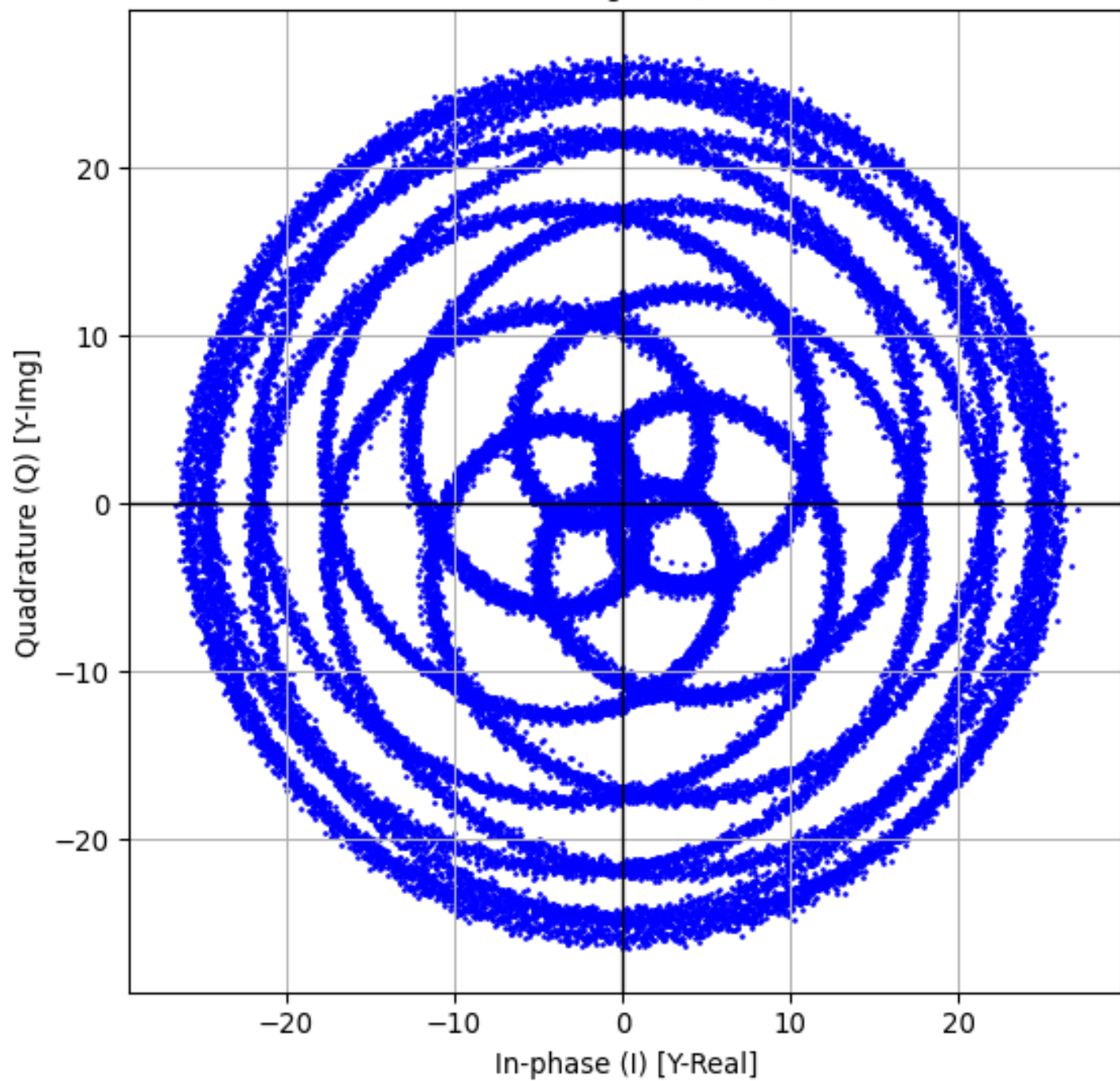
Dados Originais (Teste)



Forecast (Teste)
p:2, xl:2, r:3, i:3



Dados Originais (Total)



Forecast (Total)
p:2, xl:2, r:3, i:3

